

1. Optimize

```
List<Account> accounts = ...;
```

```
accounts.Where(...).Select(...).Where(...).OrderBy(...).GroupBy(...);
```

2. Please write code to create a GET API that will consume API - <https://api.sampleapis.com/avatar/episodes> & return the count of episodes per season in the below format as a response

3.

4. [

5. {

6. "Season": "1",

7. "EpisodeCount": 9

8. },

9. .

10. .

11. .

12.]

3. How to Handle CORS or how implement this

4. Explain microservices clean architecture

Mandatory for interface to define in program.cs file

Why program.cs

2service clas and 1 interface

How to connect to databas in .net core

Benefit of interface

What is abstract class and real time example

How to write a test case for controller

IActionResult and ActionResult

What will u choose return type for List and collection

how to communicate from Api to api .net core

2.middleware

3.Di

4.getallusers from azure ad using web services?

5.absreact and interface when use?

6.implementation of Authorization in .net core?

Collebra 2nd round

using System;

public class Program

{

 // Generic method to add numbers or concatenate strings

 public static T AddOrConcat<T>(T a, T b)

 {

 dynamic x = a;

 dynamic y = b;

 return (T)(x + y);

 }

```

public static void Main()
{
    // Numbers

    Console.WriteLine(AddOrConcat(10, 20));    // 30
    Console.WriteLine(AddOrConcat(10.5, 5.3)); // 15.8

    // Strings

    Console.WriteLine(AddOrConcat("Hello ", "World")); // Hello World
}
}

```

[22-09-2025 17:31] Muhammad Maula CNX: Write code for calling one middleware to another middleware

Example: Calling one middleware from another

```

using Microsoft.AspNetCore.Builder;
using Microsoft.AspNetCore.Http;

var builder = WebApplication.CreateBuilder(args);
var app = builder.Build();

// First middleware
app.Use(async (context, next) =>

```

```

{
    Console.WriteLine("Middleware 1: Before");
    await next(); // calls the next middleware
    Console.WriteLine("Middleware 1: After");
});

// Second middleware
app.Use(async (context, next) =>
{
    Console.WriteLine("Middleware 2: Before");
    await next(); // calls the next middleware (or endpoint)
    Console.WriteLine("Middleware 2: After");
});

// Endpoint
app.Run(async context =>
{
    Console.WriteLine("Endpoint Reached");
    await context.Response.WriteAsync("Hello World!");
});

app.Run();

```

 Output (when you hit /):

Middleware 1: Before

Middleware 2: Before

Endpoint Reached

Middleware 2: After

Middleware 1: After

👉 So to call one middleware from another, you just use:

```
await next();
```

[22-09-2025 17:33] Muhammad Maula CNX: Session is stateful or stateless? Is it mandatory?

[22-09-2025 17:34] Muhammad Maula CNX: Types of caching?

[22-09-2025 17:34] Muhammad Maula CNX: Migration from .net framework to .net core as per code perspective?

[22-09-2025 17:35] Muhammad Maula CNX: Implementation of caching?

[22-09-2025 17:35] Muhammad Maula CNX: Implementation of redis caching?

[22-09-2025 17:40] Muhammad Maula CNX: Ways to Restrict Middleware to other?

[22-09-2025 17:40] Muhammad Maula CNX: Example for singleton life time scope?

[22-09-2025 17:56] Muhammad Maula CNX: How to check or validate a token in code level

[22-09-2025 18:04] Muhammad Maula CNX: New vs override keyword ?which we use?

[22-09-2025 18:08] Muhammad Maula CNX: How to provide jwt token to the end user or non developer?

[22-09-2025 18:15] Muhammad Maula CNX: Custom middleware sample code example

[23-09-2025 11:37] Muhammad Maula CNX: Impphasis

1.The purpose of string[] args in C# Main method is:?

2.dll vs exe?check in debug folder?

[23-09-2025 11:37] Muhammad Maula CNX: 3.what is OPTIONS?

[23-09-2025 11:37] Muhammad Maula CNX: 4.private constructor

[23-09-2025 11:37] Muhammad Maula CNX: 5.constructor chaining

[23-09-2025 11:38] Muhammad Maula CNX: 6.custom middleware

[23-09-2025 11:38] Muhammad Maula CNX: 7.startup class

[23-09-2025 11:39] Muhammad Maula CNX: 8.factorial program with recursive

[23-09-2025 11:40] Muhammad Maula CNX: 9.generate substring combination from below input string

abababbbbaa

[23-09-2025 11:41] Muhammad Maula CNX: 10.reduce branch if u are static methods In unit testing

11.generic

[23-09-2025 11:50] Muhammad Maula CNX: loc container

[23-09-2025 11:52] Muhammad Maula CNX: How to send a file to the end point ?

[23-09-2025 11:53] Muhammad Maula CNX: Content negotiation?

[23-09-2025 15:37] Muhammad Maula CNX: 2.find out duplicate elements in list
`List<int> lst = new List<int>(){1,2,3,4,1,2};`

[23-09-2025 15:38] Muhammad Maula CNX: public async Task GetInfo()

{

`var departments = await getDepartment();`

`var employess = await getEmployees();`

}

any optimization required?

[23-09-2025 15:38] Muhammad Maula CNX: ===Employee===

EmpId EmpName DeptId

1 Rohan 1
2 Kalpesh 3
3 Shree 3

===Department=====

DeptId DeptName

1 Account
2 HR
3 Development

===Query Output=====

DeptId DeptName EmpCount

1 Account 1
2 HR 0
3 Development 2

[23-09-2025 15:39] Muhammad Maula CNX: Optimization for stored procedure performacne

[23-09-2025 15:40] Muhammad Maula CNX: Function vs stored procdure and Fucntiojn can support to Dml operation?

[23-09-2025 15:45] Muhammad Maula CNX: You have 1000 existing action methods in an MVC app.

The client wants to add a new action.

You want to avoid adding a new method manually every time.

Answer:Using Filters

[23-09-2025 15:46] Muhammad Maula CNX: [NonAction] attribute

[23-09-2025 15:48] Muhammad Maula CNX: In ASP.NET Core, how can you pass data from one middleware to the next and also receive data back from the next middleware during the same request? Explain with an example.

Answer:HttpContext

[23-09-2025 15:49] Muhammad Maula CNX: Advantage of Core

[23-09-2025 15:54] Muhammad Maula CNX: When we choose Service.AddScoped and Service.AddTransient?

[23-09-2025 15:58] Muhammad Maula CNX: Tortonic

1.Differnce between them

```
var temp = (from x in db.Employees
            Select x);
```

What will you have for Temp if u check in quick watch

```
var temp1 = (from x in db.Employees
            Select x).ToList();
```

What will you have for temp1 if u check in a quick watch?

[23-09-2025 18:49] +91 97423 90411: using System;

using System.Collections.Generic;

using System.Linq;

class Program

{

static void Main()

{


```
List<int> lst = new List<int>() { 1, 2, 3, 4, 1, 2 };
```

```
var duplicates = lst  
    .GroupBy(x => x)  
    .Where(g => g.Count() > 1)  
    .Select(g => g.Key)  
    .ToList();
```

```
Console.WriteLine("Duplicates: " + string.Join(", ", duplicates));
```

```
}
```

```
}
```

[23-09-2025 18:50] Muhammad Maula CNX: Var x=null;

[23-09-2025 18:51] +91 97423 90411: No we can't assign

```
Var x="somevalue";
```

```
x=null;
```

This we can we do

[23-09-2025 20:40] Muhammad Maula CNX: Those are well known answers , please respond in chat with short answers like how vijay has done,

Note: not for straight forward questions.

[24-09-2025 12:01] Muhammad Maula CNX: ♦ AddTransient (new object every time)

OTP Generator → every OTP must be fresh.

Email Formatter → each email body built separately.

♦ AddScoped (one object per request)

DbContext → same database connection for one API request.

UnitOfWork → tracks all changes in one request.

- ◆ AddSingleton (one object for whole app)

Logger → same logger instance for all requests.

Cache Service → one cache shared across all users.

[24-09-2025 12:06] Muhammad Maula CNX: // Middleware 1 - set value

```
app.Use(async (context, next) =>
```

```
{
```

```
    context.Items["UserId"] = 123; // pass data forward
```

```
    await next();
```

```
    var result = context.Items["Result"]; // read data coming back
```

```
    await context.Response.WriteAsync($"\\nResult from next: {result}");
```

```
});
```

```
// Middleware 2 - get value and return something back
```

```
app.Use(async (context, next) =>
```

```
{
```

```
    var userId = context.Items["UserId"]; // read data from previous
```

```
    context.Items["Result"] = $"Processed user {userId}";
```

```
    await next();
```

```
});
```

✅ In this flow:

Middleware 1 sets UserId.

Middleware 2 reads it, processes, and sets Result.

Middleware 1 then reads Result after next().

[24-09-2025 12:08] +91 97423 90411: var deptCount = deptList

```
.GroupJoin(empList,  
dept => dept.DeptId,  
emp => emp.DeptId,  
(dept, employees) => new  
{  
    DeptId = dept.DeptId,  
    Name = dept.Name,  
    EmpCount = employees.Count()  
});
```

```
foreach(var item in deptCount)  
{  
    Console.WriteLine($"{item.DeptId} {item.Name} {item.EmpCount}");  
}
```

[24-09-2025 12:09] Muhammad Maula CNX: SELECT

d.DeptId,

d.DeptName,

COUNT(e.EmpId) AS EmpCount

FROM

Department d

LEFT JOIN

Employee e

ON

d.DeptId = e.DeptId

GROUP BY

d.DeptId, d.DeptName

ORDER BY

d.DeptId;

[24-09-2025 12:17] Muhammad Maula CNX: public async Task GetInfo()

{

// Start both tasks concurrently

var departmentTask = getDepartment();

var employeeTask = getEmployees();

// Wait for both to complete

await Task.WhenAll(departmentTask, employeeTask);

```
// Optional: use the results if needed

var departments = await departmentTask;

var employees = await employeeTask;


// Do something with departments and employees here

}
```

Key Points:

No parameters – matches your method signature requirement.

correct me if anything wrong

Parallel execution – independent tasks run at the same time.

Proper async usage – avoids sequential awaiting.

Use the results – you can now use departments and employees inside the method.

[24-09-2025 12:20] Muhammad Maula CNX: 1. API Endpoint (Server-Side)

Use [FromForm] IFormFile for single/multiple file uploads:

```
[HttpPost("upload")]
public async Task<ActionResult> UploadFile([FromForm] IFormFile file)
{
    if (file == null || file.Length == 0)
        return BadRequest("No file uploaded.");

    var uploadPath = Path.Combine("Uploads", file.FileName);

    // Ensure directory exists
    Directory.CreateDirectory(Path.GetDirectoryName(uploadPath)!);

    // Save file efficiently
    await using var stream = new FileStream(uploadPath, FileMode.Create);
    await file.CopyToAsync(stream);

    return Ok(new { FileName = file.FileName, Size = file.Length });
}
```

Why this is best:

[FromForm] binds files from multipart/form-data.

IFormFile abstracts file handling efficiently.

Streamed copy prevents memory overload for large files.

2. Client-Side (Sending the File)

Use HttpClient with MultipartFormDataContent:

```
using var client = new HttpClient();
using var form = new MultipartFormDataContent();

var fileBytes = File.ReadAllBytes("C:\\path\\to\\file.txt");
var fileContent = new ByteArrayContent(fileBytes);
fileContent.Headers.ContentType = new
System.Net.Http.Headers.MediaTypeHeaderValue("application/octet-stream");

// 'file' must match API parameter
form.Add(fileContent, "file", "file.txt");

// Send POST request
var response = await client.PostAsync("https://localhost:5001/upload", form);
var result = await response.Content.ReadAsStringAsync();
Console.WriteLine(result);
```


Why this is best:

Uses standard multipart/form-data, compatible with browsers and other clients.

Allows adding additional parameters alongside the file.

Fully asynchronous, non-blocking.

[24-09-2025 12:41] Muhammad Maula CNX: Azure Ad configuration in appsetting file?

[24-09-2025 12:42] Muhammad Maula CNX: session is important in .NET core?

Persistent 2 round,

1.web socket

2.what middleware choose for chat application

3.cqrs pattern

4.what is the current status of concentrix

5.why do u want to join persistent

6.ado.net vs ef

7.have u created azure cloud native app

8.fabric service in azure

9.what is reverse proxy?

10.which architecture and how do u implement apis for stock applications ?

11.have u worked on azure function?which one?

True North company

1.increase performance in .net core

2.event driven for unit testing?

1.merge

2.Cte

3.when we use cte

4.only select in sql profiler?

5.tools for migration

6.scanning-static and dynamic

7.snapshot in sql

8.graphql vs rest vs grpc

9.websocket why?

10 .when we use transient and scoped

11.abstract vs interface

12.yield keyword

13.put vs patch

14.sso implementation using octa

15.csp

16.api communication

17.authentication

18.code smell

19.status code

Dbcobtext and dbset

- 1.shell script in ef core
- 2.azure ad configuration in app settings
- 3.how to mock for controller or service in xunit framework
- 4.how to load postgresql and sql server in .net core (when 2 Dbs)
- 5.how to validate jwt token in .net core
- 6.how to implement logging

How to generate a certificate file if https is blocking not proper configure ssl and TLS